NanoparticleAtomCounter: A Python/Web-based package to convert nanoparticle geometry from Transmission Electron Microscopy into atom counts

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Abstract

NanoparticleAtomCounter converts transmission electron microscopy (TEM) derived geometry into atom counts for supported nanoparticles. Using the radius and contact angle of a nanoparticle, *NanoparticleAtomCounter* models nanoparticles as spherical caps and analytically estimates the total, surface, interfacial, and perimeter atoms without any atomistic modelling, allowing the conversion of hundreds of thousands of geometries into atom counts in seconds.

Keywords

Nanoparticle, Transmission Electron Microscopy, Spherical cap, Catalysis

Significance

Supported nanoparticles are important catalysts, and central to their reactivity is the binding site, the local environment where reactants adsorb. Quantifying how many binding sites exist and where they sit is essential for understanding catalyst performance. While imaging techniques such as transmission electron microscopy (TEM) give precise shapes and sizes, they do not directly yield atomic-level counts across different regions of the nanoparticle. Converting the size and shape descriptors into atom counts is critical, as atoms located at interfaces, perimeters, and outer surfaces often exhibit different reactivity. Existing approaches to determine atom counts include the construction of full atomistic models for each particle geometry, an approach that scales poorly and is impractical for high-throughput TEM image screening.

Our package, *NanoparticleAtomCounter*, addresses the challenge of estimating atom counts in nanoparticles by using only two inputs: radius and contact angle, both of which can be obtained from TEM imaging. *NanoparticleAtomCounter* approximates nanoparticle morphology as a spherical cap and uses analytical geometry to estimate the atom counts in the interface, perimeter, and outer surface without requiring atomic-resolution modeling. Hundreds of thousands of calculations can be completed in a few seconds. Thus, researchers can translate TEM-derived geometry into quantitative binding site populations, enabling turnover frequencies to be reported on a per-site basis, rather than per-mass/area.

Software Description

Below, we describe the methods used to calculate the number of total, interfacial, perimeter, and surface atoms of supported nanoparticles (Figure 1). We assume a monometallic nanoparticle shaped as a spherical cap of footprint radius r, radius of curvature R, and contact angle θ. Here, we illustrate only a case with $\theta > 90$; similar equations (with slight modifications) apply when $\theta \leq 90$.

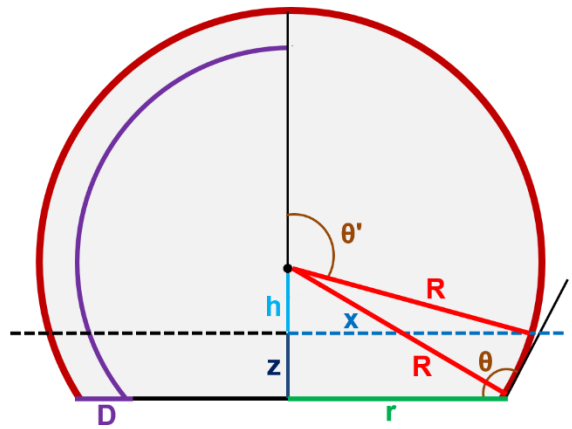

Figure 1. Spherical cap model of a supported nanoparticle with $\theta > 90$.

Interfacial count

We first consider the *entire* interfacial region (i.e. interface + perimeter) of thickness z (exaggerated in Figure 1), which is equal to the planar spacing at the nanoparticle-support interface. We calculate the planar spacing using the atomic simulation environment (ASE)[1] for all facets from {0,0,1} till {3,3,3}. The volume for the entire interfacial region is given by the formula for the volume of a spherical segment:

$$V_{entire\ interface} = \frac{\pi}{6} z(3x^2 + 3r^2 + z^2) \quad (1)$$

where x is the footprint radius at height z above the nanoparticle-support interface. The only unknown term is x, which is expressed by Pythagoras' theorem:

$$x^2 = R^2 - h^2 \quad (2)$$

where R is given as:

$$R = \frac{r}{\sin\theta} \quad (3)$$

and h is obtained from R by Pythagoras' theorem:

$$R^2 = r^2 + (h + z)^2 \quad (4)$$

Thus, x is given as:

$$x^2 = r^2 + (h + z)^2 - h^2 \quad (5)$$

We can then convert the entire interfacial volume $V_{entire\ interface}$ to the number of interfacial + perimeter atoms $N_{entire\ interface}$:

$$N_{entire\ interface} = \frac{V_{entire\ interface}}{V_M} \times N_A \quad (6)$$

where N_A is Avogadro's number and V_M is the molar volume, calculated with ASE.

A faster method, which does not consider θ and hence assumes a negligible thickness of the interfacial region, calculates the number of interfacial atoms thus:

$$N_{entire\ interface} = \sigma_{interface} \times A_{entire\ interface} \quad (7)$$

where $\sigma_{interface}$ is the atomic surface density (atoms per unit area) at the interface and $A_{entire\ interface}$ is the area of the entire interface. $A_{entire\ interface}$ is calculated by:

$$A_{entire\ interface} = \pi r^2 \quad (8)$$

and $\sigma_{interface}$, thus:

$$\sigma_{interface} = \frac{z \times N_A}{V_M} \quad (9)$$

Lastly, as what we have called the "entire interface" is comprised of the interface and the perimeter, we have the number of interfacial atoms, $N_{interface}$ as:

$$N_{interface} = N_{entire\ interface} - N_{perimeter} \quad (10)$$

The procedure for estimating $N_{perimeter}$ is described below.

Perimeter count

We assume that the entire interface is an annular ring of thickness D, which is the diameter of an atom of the nanoparticle and therefore the thickness of the perimeter. Thus, the radius of the interface is r-D. The volume of the interface $V_{interface}$ is therefore calculated with equation 1, where r-D is substituted for r. Note that before applying equation 1, x is also recalculated using equation 5, with r-D substituted for r.

The perimeter's volume $V_{perimeter}$ is:

$$V_{perimeter} = V_{entire\ interface} - V_{interface} \quad (11)$$

$N_{perimeter}$ is then calculated using equation 6, with $V_{perimeter}$ substituted for $V_{entire\ interface}$.

Using the area-based method, we can calculate $N_{perimeter}$ using equations 7-9, but with equation 8 modified to:

$$A_{perimeter} = \pi r^2 - \pi (r - D)^2 \quad (12)$$

Total count

The total volume V_{total} of the nanoparticle uses the formula:

$$V_{total} = \frac{\pi \times r^3 \times \alpha(\theta) \times \beta(\theta)}{3} \quad (13)$$

where:

$$\alpha(\theta) = \frac{1}{1 + \cos\theta} \quad (14)$$

and:

$$\beta(\theta) = \frac{(2+\cos\theta)(1-\cos\theta)}{\sin\theta} \tag{15}$$

The total number of atoms N_{total} is then calculated using equation 6, with N_{total} substituted for $N_{entire\ interface}$

Surface count

From Figure 1:

$$\cos(180-\theta) = -\cos\theta = \frac{h+z}{R} \tag{16}$$

Which gives:

$$h + z = -R\cos\theta \tag{17}$$

Similarly:

$$\cos(180-\theta') = -\cos\theta' = \frac{h}{R} \tag{18}$$

Which gives:

$$h = -R\cos\theta' \tag{19}$$

Therefore:

$$h \equiv (h+z) - z = -R\cos\theta - z = -R\cos\theta' \tag{20}$$

Dividing through by -*R* gives:

$$\cos\theta' = \frac{h}{-R} = \frac{z}{R} + \cos\theta \tag{21}$$

Which gives us the contact angle at height *z* (i.e. *θ'*). The surface area of the region of the nanoparticle above *z* (i.e. the surface area excluding the area of the perimeter) is calculated with the formula for the curved surface area of a spherical cap:

$$A_{curved\ surface} = 2\pi r^2 \alpha(\theta') \tag{22}$$

The atomic density at the surface is calculated similarly to equation 9:

$$\sigma_{surface} = \frac{z_{surface} \times N_A}{V_M} \tag{23}$$

where $z_{surface}$ is the planar spacing at the nanoparticle's outer surface.

Finally, the number of surface atoms (*excluding* the perimeter atoms) is calculated with equation 7, substituting $A_{curved\ surface}$ for $A_{entire\ interface}$ and $\sigma_{surface}$ for $\sigma_{interface}$.

Illustrative Examples

NanoparticleAtomCounter is available both as a web app and as a command line tool. In this section, we provide an example of how to use in both modes

The user is first required to prepare an input csv file having the following headers:

r (A), R(A), Theta, Element, Interface Facet, Surface Facet

where:

r (A): nanoparticle footprint radius, in Å, e.g: 9.8.

R (A): nanoparticle radius of curvature, in Å, e.g: 19.8. Only "r (A)" or "R (A)" need be supplied, as one can be calculated from the other (equation 3). Therefore, if both "r (A)" and "R (A)" are supplied, only "r (A)" is used.

Theta: contact angle, in °, e.g: 77.

Element: what element the nanoparticle is comprised of, e.g: Sc.

Interface Facet: The type of facet at the nanoparticle-support interface, e.g. (1,0,0). If not supplied, the most stable facet will be used.

Surface Facet: The type of facet at the nanoparticle-gas interface, e.g. (1,1,1). If not supplied, the most stable facet will be used.

Calculations with the web app

Figure 2 shows the web app, which is hosted at https://nanoparticleatomcounter.streamlit.app/. In the side bar, a sample input file can be downloaded.

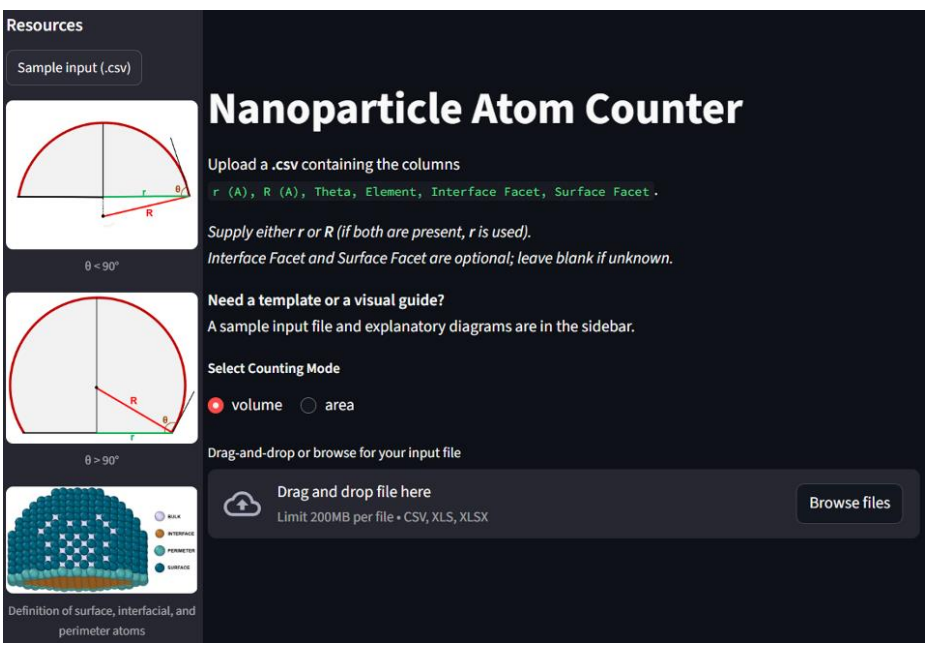

Figure 2. Web application interface for *NanoparticleAtomCounter*

After uploading their input csv, the user then selects the mode for counting: "volume" (equations 1-6, 11, 13-23), or "area" (equations 7-9, 12, 13-23). "volume" is selected by default. A "Run calculation" button appears. After clicking the button, the results are displayed in a table having the headers:

Perimeter, Interface, Surface, Total

with the estimated number of atoms displayed beneath each header. Figure 3 clarifies what we refer to as "perimeter", "interface", and "surface":

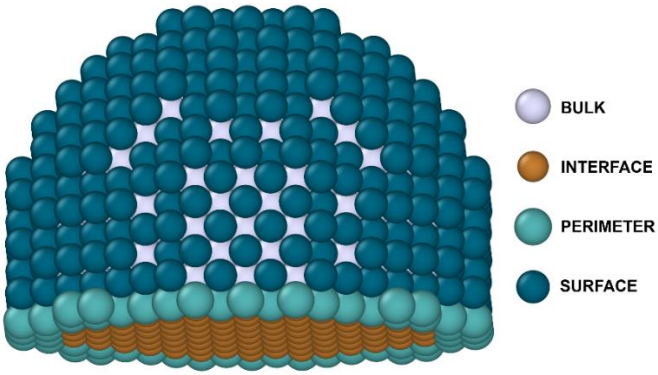

Figure 3. Definition of perimeter, interface, and surface atoms

A "Download CSV" button can also be clicked to download the results as a CSV.

Calculations with the command line interface

NanoparticleAtomCounter can also be used as a command line tool. The source code and installation instructions are at https://github.com/szilvasi-group/NanoparticleAtomCounter.

After preparing the input CSV (see instructions above), *NanoparticleAtomCounter* is run thus:

nanoparticle-atom-count -i input_name.csv -o output_name.csv -m counting_mode

counting_mode should be "volume" or "area". If not specified, "volume" will be used.

On running the above command, the results file, *output_name.csv*, is created.

Software Testing and Benchmarks

Computational Efficiency

NanoparticleAtomCounter demonstrates high computational efficiency, processing hundreds of structures within milliseconds and scaling to one million structures in just over a minute (Figure 4).

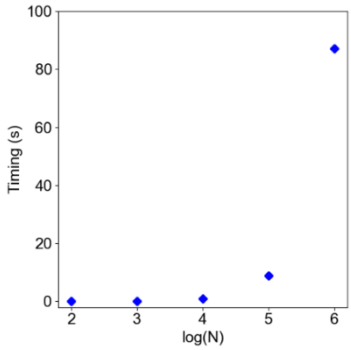

Figure 4. Speed of calculation as a function of number of structures processed (*N*)

Accuracy

To benchmark the accuracy of *NanoparticleAtomCounter,* we compare its predictions to the atom counts obtained from fully atomistic models of nanoparticles. The dataset we use consists of 64 spherical cap-like Ag nanoparticles with contact angles ranging from 30-160 ° and radii of curvature from 10-50 Å. We generate atomistic approximate spherical caps in ASE by constructing bulk FCC Ag and deleting atoms lying beyond a specified radius from the center of the simulation cell.

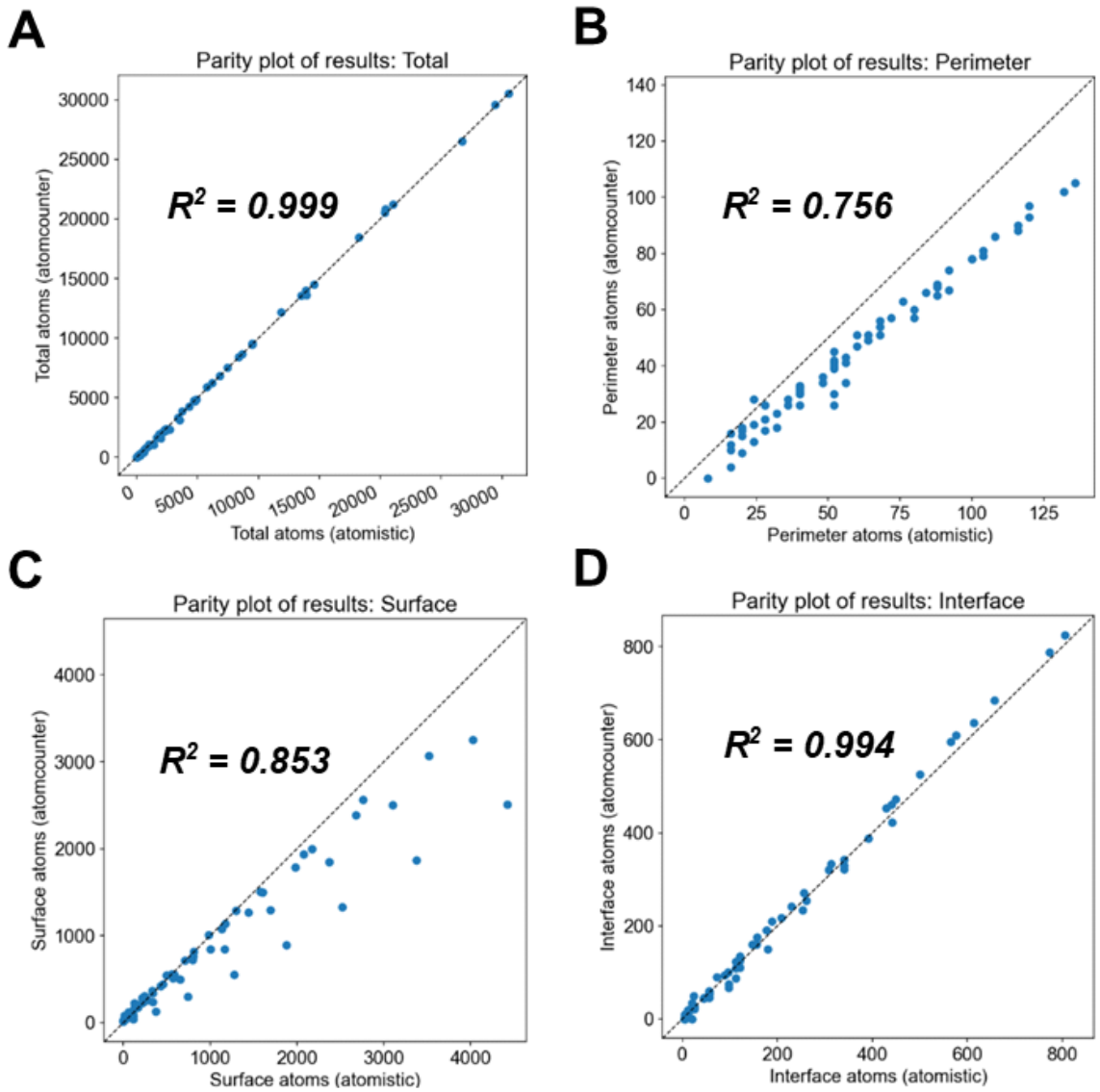

Figure 5. Parity plots of atom counts using *NanoparticleAtomCounter* vs a fully atomistic model: (A) total atoms (B) perimeter atoms (C) surface atoms (D) interfacial atoms

Figure 5 presents parity plots comparing atom counts for the different nanoparticle regions defined in Figure 3, demonstrating generally strong agreement between the fully atomistic models and

NanoparticleAtomCounter. From the results plotted in Figure 5, Figure 6 is constructed by evaluating the absolute percent difference:

$$\text{absolute percent difference} = \left|\frac{100 * (N_{atomistic} - N_{NanoparticleAtomCounter})}{N_{atomistic}}\right| \quad (24)$$

where $N_{atomistic}$ and $N_{NanoparticleAtomCounter}$ refer to the atom counts according to the atomistic model and the *NanoparticleAtomCounter*, respectively.

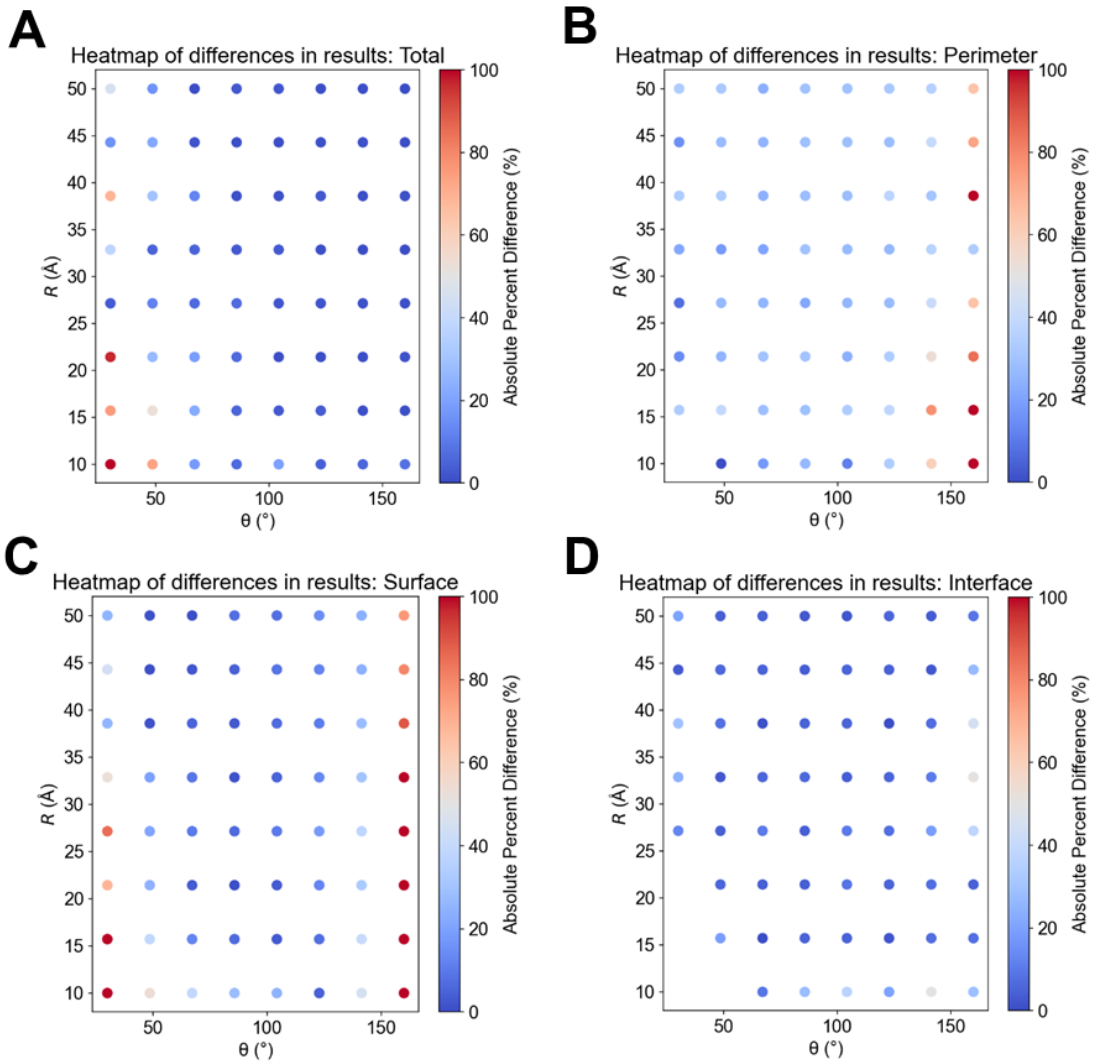

Figure 6. Heatmap of the absolute percent difference of atom counts using *NanoparticleAtomCounter* vs a fully atomistic model: (A) total atoms (B) perimeter atoms (C) surface atoms (D) interfacial atoms

Figure 6 shows that, overall, *NanoparticleAtomCounter* reproduces atomistic atom counts with good accuracy. The largest deviations (reddish points) typically occur for nanoparticles with small radius of curvature (e.g. R < 20 Å). The sometimes large deviations in surface atom counts arise from the use of a surface atomic density based on a single dominant facet without accounting for the coexistence of other facets, as well as from representing the surface region as a simple geometric shell. Deviations in perimeter atom counts primarily originate from modeling the perimeter as an annular region; additionally, at large

contact angles (e.g. $\theta > 150$ °), where the number of perimeter atoms becomes small, so that even modest absolute differences translate into large percentage errors.

Availability

Version 0.1.8 of *NanoparticleAtomCounter* is freely available under the MIT License. The source code, tests, examples, and documentation are hosted at https://github.com/szilvasi-group/NanoparticleAtomCounter. The web version is hosted at https://nanoparticleatomcounter.streamlit.app/

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References

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